```
clc;
clear;
close all;

% EXPERIMENT 7: FIR FILTER DESIGN
% Parks-McClellan and Frequency-Sampling Methods

% 1. Parameters
Fs = 10000;
N = 40;
L = N + 1;

fp = 1500;
fsb = 2000;

% 2. Parks-McClellan FIR Filter Design
f_edges = [0 fp fsb Fs/2] / (Fs/2);
a_edges = [1 1 0 0];
weights = [1 10];

b_pm = firpm(N, f_edges, a_edges, weights);

% 3. Frequency-Sampling FIR Filter Design
% Generate equally spaced frequency samples

k = 0:N;
f_samples = min(k, L-k) * Fs/L;

% Desired magnitude at each frequency sample
A = interp1([0 fp fsb Fs/2], ...
            [1 1 0 0], f_samples, 'linear');

% Enforce symmetry
for i = 1:L
    j = L-i+1;
    A(j) = A(i);
end

% Add linear-phase delay
H_samples = A .* exp(-1j*2*pi*k*(N/2)/L);

% Calculate FIR coefficients
b_fs = real(ifft(H_samples));

% 4. Calculate Frequency Responses
nfft = 4096;

H1 = fft(b_pm, nfft);
H2 = fft(b_fs, nfft);

H1 = H1(1:nfft/2+1);
H2 = H2(1:nfft/2+1);
```

```
f = (0:nfft/2) * Fs/nfft;

% Magnitude in dB
Mag1 = 20*log10(max(abs(H1), 1e-6));
Mag2 = 20*log10(max(abs(H2), 1e-6));

% Phase in degrees
Phase1 = unwrap(angle(H1))*180/pi;
Phase2 = unwrap(angle(H2))*180/pi;

% 5. Display Coefficients
disp('PARKS-MCCLELLAN FILTER COEFFICIENTS');
disp(b_pm(1:10));

disp('FREQUENCY-SAMPLING FILTER COEFFICIENTS');
disp(b_fs(1:10));

% 6. OUTPUT GRAPH 1: MAGNITUDE RESPONSE
figure(1);

plot(f, Mag1, 'b', 'LineWidth', 1.5);
hold on;
plot(f, Mag2, 'r--', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('FIR Filter Magnitude Response Comparison (Order N = 40)');

legend('Parks-McClellan', 'Frequency-Sampling', ...
       'Location', 'southwest');

xlim([0 Fs/2]);
ylim([-100 10]);

% 7. OUTPUT GRAPH 2: PHASE RESPONSE
figure(2);

plot(f, Phase1, 'b', 'LineWidth', 1.5);
hold on;
plot(f, Phase2, 'r--', 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('FIR Filter Phase Response Comparison');

legend('Parks-McClellan', 'Frequency-Sampling', ...
       'Location', 'best');

xlim([0 Fs/2]);
```

```
% 8. Completion Message
disp('Both output graphs have been generated.');
```

```
PARKS-MCCLELLAN FILTER COEFFICIENTS
  Columns 1 through 7

    0.0034    0.0030   -0.0016   -0.0090   -0.0121   -0.0055    0.0072

  Columns 8 through 10

    0.0136    0.0041   -0.0148

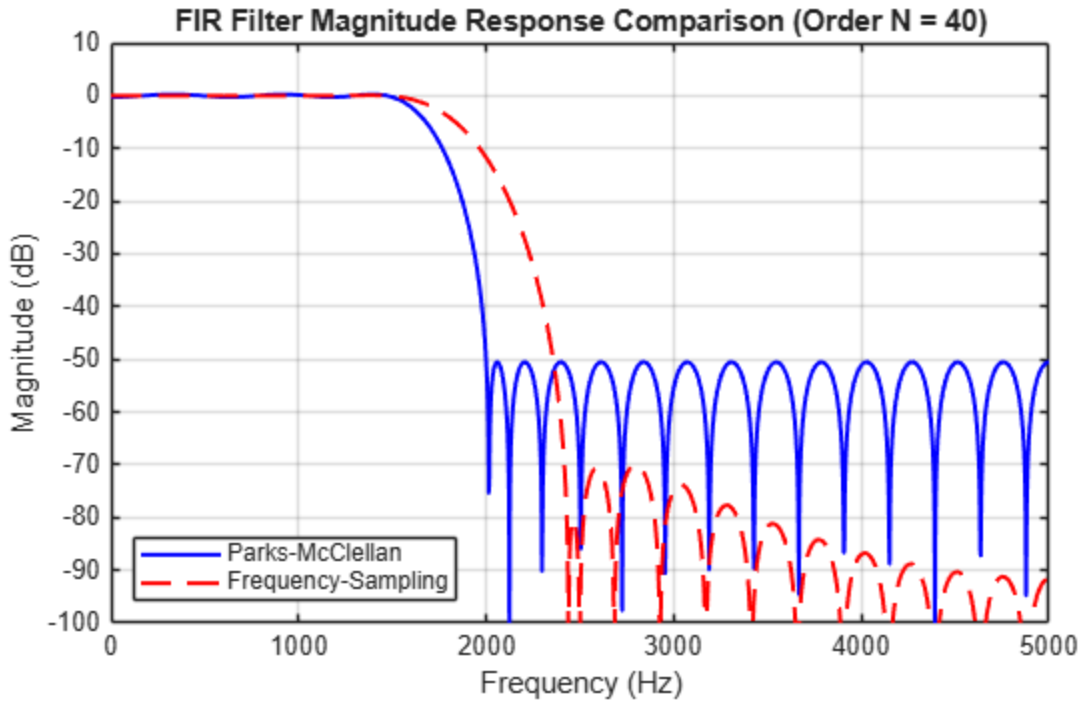
FREQUENCY-SAMPLING FILTER COEFFICIENTS
  Columns 1 through 7

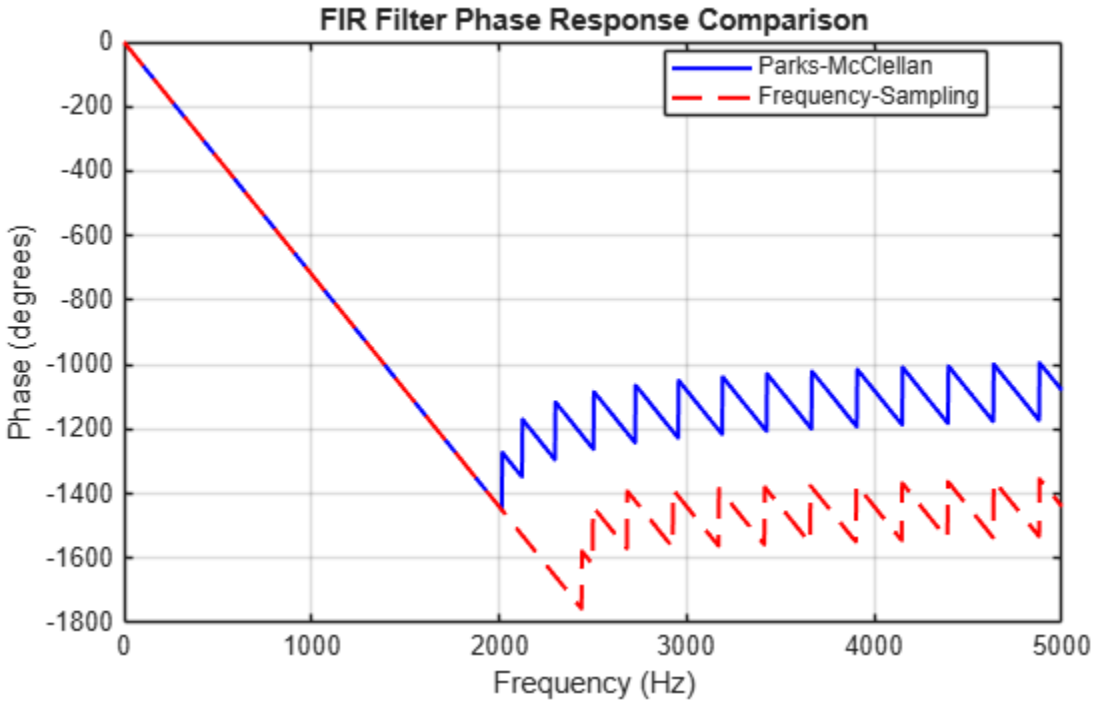
   -0.0000    0.0001    0.0006    0.0007   -0.0007   -0.0030   -0.0024

  Columns 8 through 10

    0.0034    0.0086    0.0033

Both output graphs have been generated.
```






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